APPENDIX II. DEBT SUSTAINABILITY ANALYSIS (UPDATE)

This appendix updates the joint Bank/Fund debt sustainability analysis (DSA) from June 2009. Compared to the previous results, the debt sustainability outlook worsens over the medium term because of faster debt accumulation in the short term, but improves in the long term with continued fiscal consolidation efforts. Burkina Faso is still classified as a medium performer and the country remains at high risk of debt distress.

I. EXTERNAL DEBT SUSTAINABILITY ANALYSIS

A. What Has Changed Since the Last DSA?

1. **The medium-term economic outlook remains broadly unchanged**. Real GDP growth for 2009 has been lowered by 0.4 percentage points because of the greater impact of the global economic downturn, but projections remain unchanged over the medium term (Box 1). Real GDP growth is projected to average 6 percent in the long run, in line with the average for the past ten years. Export diversification and structural reforms implementation remain key to sustain growth over the medium term (Figure 1).

Box 1. Burkina Faso—Macroeconomic Assumptions

Real GDP growth is projected to average 6 percent in the medium and long run. Growth in 2009 and 2010 is projected to be below trend as a result of exogenous shocks (global economic crisis in 2008-09 and heavy flooding in September 2009). Over the medium and long term, growth is expected to reach 6 percent, as a result of improvements in productivity, enhanced infrastructure, and export diversification.

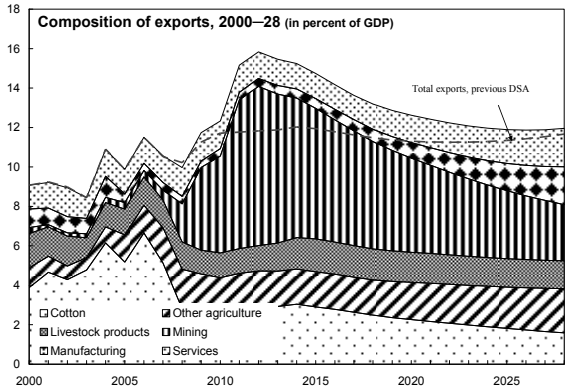
Inflation is expected to average about 2 percent over the medium and long term.

The overall fiscal deficit, with and without grants, is projected to decrease over time as revenue increases towards the WAEMU tax revenue target of 17 percent of GDP. The revenue increase is supported by ongoing efforts to strengthen tax and customs administration, and comprehensive tax policy reforms. Expenditure is projected to remain high to meet the authorities' PRSP objectives. External financing requirements are to be met equally by grants and loans, a conservative assumption based on recent experience.

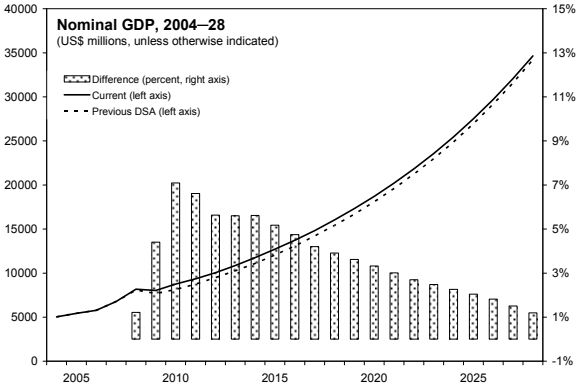
The current account deficit is expected to narrow in the medium and long run as the tighter fiscal stance dampens import demand and gold production contributes to an increase in exports.

Appendix II—Figure 1. Burkina Faso: Medium-Term Framework

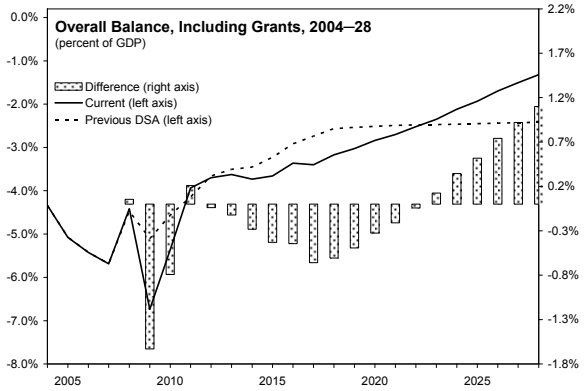
Exports are expected to be more diversified over the medium term.

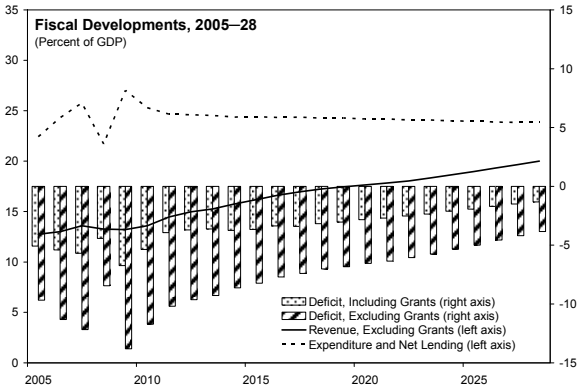


Nominal GDP is projected to be slightly higher, largely because of price developments.

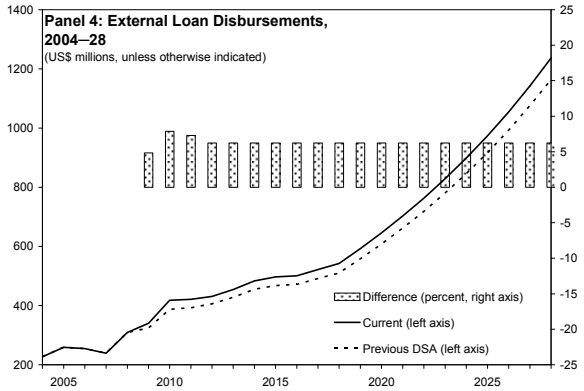


Following a deterioration in the short term linked to exogenous shocks, the implementation of fiscal reforms will contribute to reducing the overall fiscal deficit over the medium term.

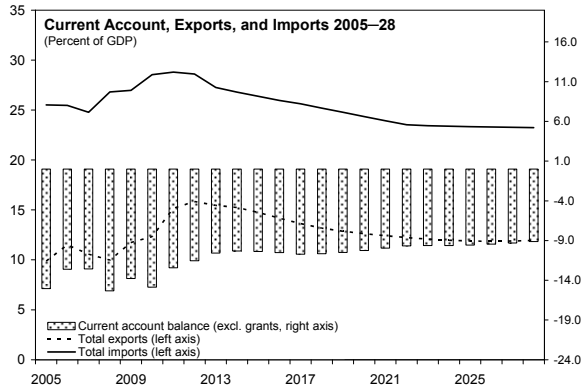




High external borrowing over the medium term underscores the importance of concessional financing.



Current account balance is expected to narrow as tighter fiscal stance dampens imports demand.



Sources: Burkinabè authorities, World Economic Outlook, and IMF staff estimates and projections.

2. **Fiscal developments reflect the impact of recent shocks to the Burkinabè economy.** The deterioration in the overall fiscal deficit in 2009 and 2010 leads to an increase in the NPV of debt-to-revenue and NPV of debt-to-GDP ratios from 2011 onwards, but the impact of the deterioration diminishes over the medium and the long term because of fiscal consolidation efforts, which will notably benefit from higher revenue linked to ongoing tax policy reform and planned measures to control current expenditure.

B. Assessing Vulnerabilities

3. **Stress tests show that a worse-than-expected export performance and less favorable financing conditions pose the greatest risks to the external debt outlook**. Burkina Faso remains heavily dependent on cotton and gold exports. Therefore, downside risks are related to fluctuations in production and terms of trade shocks. This underscores the importance for the authorities to secure sufficiently concessional financing.

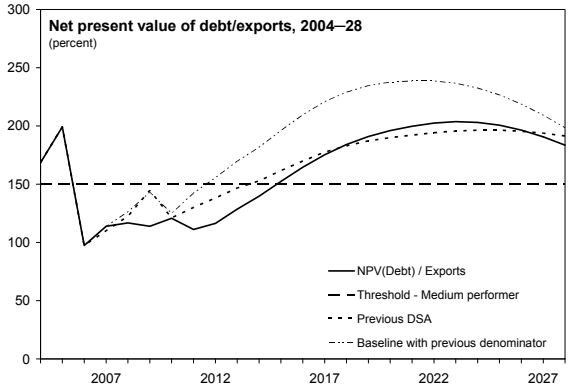
4. **A continuous decline in the fiscal deficit is key to ensure that the rate of external debt accumulation remains manageable**. Following an increase in debt accumulation in the short term, which contributes to a worsening of revenue and GDP debt sustainability framework (DSF) indicators over the medium term, sustained fiscal consolidation efforts would lead to significantly lower debt accumulation and a clear improvement in debt indicators over the long run (Figure 2).

C. Risk of Debt Distress

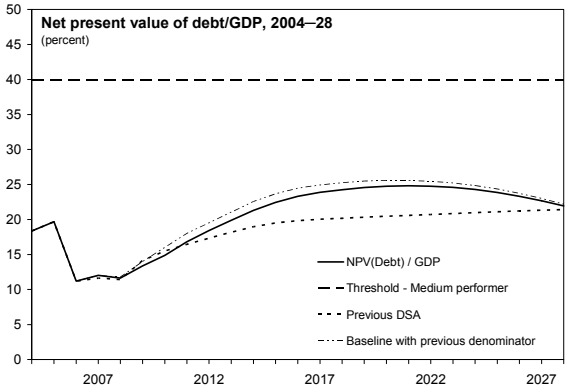
5. **The risk of debt distress is high because the threshold for the NPV of debt to exports ratio is breached**. The peak ratio of about 203 percent in the baseline projection of the NPV of debt-to-exports ratio is well above the 150 percent threshold, warranting a classification of high risk of debt distress under DSF guidelines. The GDP- and revenue-based DSF indicators remain below their indicative thresholds, including under the alternative scenarios and stress tests.

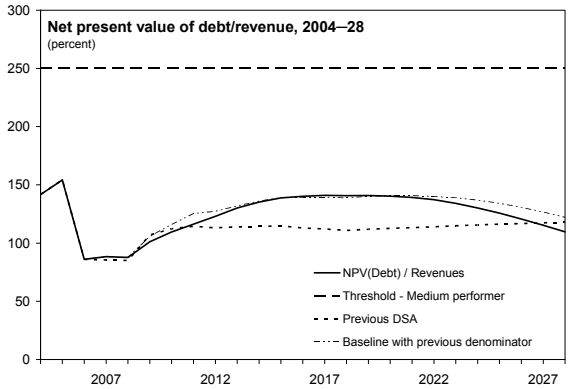
Appendix II—Figure 2. Burkina Faso: Current vs. Previous DSA[3]

The main risk remains the NPV of debt to exports ratio.

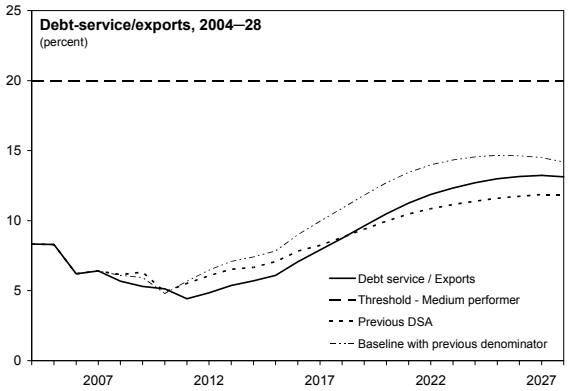


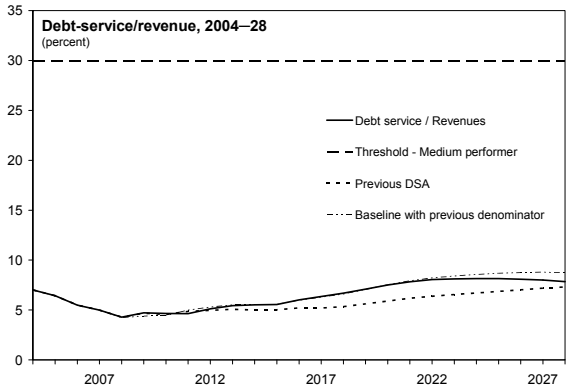
NPV of debt to GDP and revenue ratios increase over the medium term but remain below the indicative thresholds.





Debt service ratios are well below thresholds.





Sources: Burkinabè authorities, World Economic Outlook, and IMF staff estimates and projections.

[3] Previous DSA refers to the June 2009 joint Bank/Fund DSA.

II. Public Debt Sustainability Analysis

6. **The results for the fiscal DSA are similar to those of the external DSA**. The domestic debt stock is low. Gross domestic debt is estimated at about 2.7 percent of GDP at end-2008, and the net debt at 1.3 percent of GDP. Although domestic debt increases in 2009 and 2010 because of increased financing needs triggered by exogenous shocks, it is assumed to decline over the projection period as the authorities seek to avoid more costly domestic financing of the fiscal deficit. Public debt dynamics are thus largely determined by the evolution of external debt. Stress tests and alternative scenarios show that failure to reduce the current fiscal deficits would lead to ever-increasing debt indicators, demonstrating the importance of fiscal consolidation efforts.

III. Conclusion

7. **Burkina Faso's risk of debt distress is high** on account of the NPV of debt-to-exports ratio; all other debt indicators remain comfortably below their policy-dependent thresholds under the baseline scenario. Nevertheless, the high risk of debt distress and the accumulation of debt as a result of recent shocks underscore the importance of limiting external borrowing to concessional loans, implementing a prudent fiscal policy, and making continued efforts to diversify and increase exports. Moreover, the authorities should continue to improve their policy and institutional environment, which over time may result in a higher CPIA rating and higher debt thresholds.

Appendix II—Table 1. Burkina Faso: External Debt Sustainability Framework, Baseline Scenario, 2008–28 1/

(percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2005	2006	2007			2008	2009	2010	2011	2012	2013	2008–13 Average	2018	2028	2014–28 Average
External debt (nominal) 1/	**38.9**	**21.1**	**19.8**			**19.8**	**20.6**	**22.2**	**24.4**	**26.2**	**27.8**		**33.1**	**25.3**	
o/w public and publicly guaranteed (PPG)	38.9	21.1	19.8			19.8	20.6	22.2	24.4	26.2	27.8		33.1	25.3	
Change in external debt	-2.9	-17.8	-1.3			0.0	0.9	1.6	2.2	1.9	1.6		0.2	-1.4	
Identified net debt–creating flows	8.0	6.7	0.1			10.0	8.0	8.8	8.0	7.5	6.5		6.7	6.2	
Non–interest current account deficit	**11.4**	**9.3**	**8.1**	**10.0**	**1.2**	**11.7**	**8.9**	**10.3**	**9.5**	**8.7**	**7.8**		**8.2**	**7.4**	7.7
Deficit in balance of goods and services	15.6	14.0	14.2			16.8	15.2	16.2	13.6	12.8	11.8		12.0	11.3	
Exports	9.9	11.5	10.6			10.0	11.7	12.3	15.2	15.8	15.5		13.2	12.0	
Imports	25.5	25.5	24.8			26.8	27.0	28.5	28.8	28.6	27.3		25.2	23.2	
Net current transfers (negative = inflow)	-4.3	-4.4	-6.0	-4.8	0.9	-5.0	-6.2	-5.8	-4.0	-4.0	-3.9		-3.7	-3.9	-3.8
o/w official	-3.4	-3.0	-4.3			-3.5	-4.7	-4.4	-2.7	-2.6	-2.4		-2.0	-1.3	
Other current account flows (negative = net inflow)	0.1	-0.2	-0.2			-0.2	-0.1	-0.1	-0.1	-0.1	-0.1		-0.1	0.0	
Net FDI (negative = inflow)	**-0.6**	**-0.6**	**-5.1**	**-0.9**	**1.5**	**-1.0**	**-0.4**	**-0.9**	**-0.6**	**-0.2**	**-0.2**		**-0.2**	**-0.2**	-0.2
Endogenous debt dynamics 2/	**-2.8**	**-2.0**	**-2.9**			**-0.7**	**-0.4**	**-0.6**	**-0.9**	**-1.1**	**-1.1**		**-1.3**	**-1.0**	
Contribution from nominal interest rate	0.3	0.2	0.2			0.2	0.2	0.2	0.2	0.3	0.3		0.5	0.5	
Contribution from real GDP growth	-2.8	-2.0	-0.6			-0.9	-0.6	-0.8	-1.1	-1.4	-1.5		-1.8	-1.5	
Contribution from price and exchange rate changes	-0.3	-0.3	-2.4			…	…	…	…	…	…		…	…	
Residual (3–4) 3/	**-10.9**	**-24.6**	**-1.4**			**-10.0**	**-7.2**	**-7.2**	**-5.9**	**-5.6**	**-4.9**		**-6.5**	**-7.6**	
o/w exceptional financing	0.0	-23.6	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	19.7	11.2	12.0			11.6	13.4	14.9	16.8	18.4	19.9		24.3	21.9	
percent of exports	199.1	97.3	113.8			116.7	113.9	120.6	111.0	116.4	128.6		184.0	183.5	
NPV of PPG external debt	**19.7**	**11.2**	**12.0**			**11.6**	**13.4**	**14.9**	**16.8**	**18.4**	**19.9**		**24.3**	**21.9**	
percent of exports	**199.1**	**97.3**	**113.8**			**116.7**	**113.9**	**120.6**	**111.0**	**116.4**	**128.6**		**184.0**	**183.5**	
percent of government revenues	**154.2**	**86.1**	**88.5**			**87.7**	**101.2**	**109.4**	**116.3**	**123.1**	**130.2**		**140.7**	**109.6**	
Debt service-to-exports ratio (percent)	**8.3**	**6.2**	**6.4**			**5.7**	**5.3**	**5.1**	**4.4**	**4.8**	**5.4**		**8.7**	**13.1**	
PPG debt service-to-exports ratio (percent)	**8.3**	**6.2**	**6.4**			**5.7**	**5.3**	**5.1**	**4.4**	**4.8**	**5.4**		**8.7**	**13.1**	
PPG debt service-to-revenue ratio (percent)	**6.4**	**5.5**	**5.0**			**4.3**	**4.7**	**4.7**	**4.6**	**5.1**	**5.4**		**6.7**	**7.8**	
Total gross financing need (US$ millions)	631.1	548.6	248.7			918.0	728.6	878.6	897.3	934.5	919.6		1474.5	3046.5	
Non–interest current account deficit that stabilizes debt ratio	14.3	27.2	9.4			11.6	8.0	8.8	7.3	6.8	6.2		8.0	8.8	
Key macroeconomic assumptions															
Real GDP growth (percent)	7.1	5.5	3.6	5.6	1.9	5.2	3.1	4.2	5.3	6.0	6.0	5.0	6.0	5.9	5.9
GDP deflator in US$ terms (percent change)	0.8	0.7	13.1	5.3	10.3	14.9	-4.8	4.6	1.6	1.3	2.0	3.2	2.0	2.0	2.0
Effective interest rate (percent) 5/	0.7	0.7	1.2	0.6	0.4	1.3	1.0	0.8	1.1	1.3	1.3	1.1	1.5	1.8	1.6
Growth of exports of G&S (US$ terms, percent)	-2.1	23.7	7.7	12.4	22.2	14.1	15.5	14.4	31.5	12.1	5.5	15.5	4.6	8.6	6.2
Growth of imports of G&S (US$ terms, percent)	11.7	6.1	14.0	11.1	12.8	30.8	-1.3	15.4	7.9	6.6	3.0	10.4	6.3	7.8	6.9
Grant element of new public sector borrowing (percent)	…	…	…	…	…	47.7	51.5	44.0	41.9	41.9	41.2	44.7	40.8	40.0	40.0
Aid flows (US$ billions) 7/	554.8	541.6	678.6			588.2	886.1	922.6	1062.6	1084.4	1139.9		1234.6	1745.3	
o/w Grants	249.8	257.2	439.4			329.6	567.6	555.9	585.4	595.6	613.1		617.3	872.6	
o/w Concessional loans	259.4	254.9	239.2			258.6	318.5	366.7	477.2	488.8	526.9		617.3	872.6	
Grant–equivalent financing (percent of GDP) 8/	…	…	…			5.7	8.8	8.4	8.4	8.0	7.6		5.4	3.5	5.0
Grant–equivalent financing (percent of external financing) 8/	…	…	…			76.0	84.6	76.4	73.9	73.8	72.8		70.6	70.4	70.2
Memorandum items:															
Nominal GDP (US$ millions)	5436.9	5776.6	6768.8			8184.4	8031.1	8751.6	9362.5	10048.7	10863.3		16033.8	34632.8	
(NPVt–NPVt–1)/GDPt–1 (percent)						2.0	1.5	2.8	3.1	2.9	3.1	2.6	2.3	1.0	2.0

Source: Staff simulations.

1/ Includes both public and private sector external debt.

2/ Derived as [r – g – r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections, capital transfers are included–in particular project grants, which are projected to average about 2.5 percent of GDP over the long term, and about 4 percent over 2008-12 due to MCC grants—and private, non-debt-creating capital inflows. Projections also includes contribution from price and exchange rate changes.

4/ Assumes that NPV of private sector debt is equivalent to its face value.

5/ Current–year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant–equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the NPV of new debt).

Appendix II—Table 2. Burkina Faso: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28
(Percent)

	Projections							
	2008	2009	2010	2011	2012	2013	**2018**	2028
NPV of debt-to-GDP ratio								
Baseline	12	13	15	17	18	20	**24**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	12	12	13	14	16	18	**23**	21
A2. New public sector loans on less favorable terms in 2008–27 2/	12	14	17	20	23	26	**34**	35
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	12	13	15	17	18	20	**24**	22
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	12	15	20	22	23	24	**27**	22
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	12	13	16	19	20	22	**27**	24
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	12	16	19	21	23	24	**27**	22
B5. Combination of B1–B4 using one–half standard deviation shocks	12	14	19	21	22	23	**26**	22
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	12	20	22	25	27	29	**35**	32
NPV of debt-to-exports ratio								
Baseline	117	114	121	111	116	129	**184**	183
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 1/	117	103	105	95	99	114	**176**	175
A2. New public sector loans on less favorable terms in 2007–26 2/	117	122	139	134	146	166	**259**	294
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	117	114	121	111	116	129	**184**	183
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	117	165	259	231	236	255	**332**	306
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	117	114	121	111	116	129	**184**	183
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	117	132	156	139	142	154	**202**	187
B5. Combination of B1–B4 using one–half standard deviation shocks	117	150	204	182	186	201	**262**	242
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	117	114	121	111	116	129	**184**	183
NPV of debt-to-revenue ratio								
Baseline	88	101	109	116	123	130	**141**	110
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007–26 1/	88	91	95	99	105	115	**134**	104
A2. New public sector loans on less favorable terms in 2007–26 2/	88	109	126	140	154	168	**198**	176
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	88	101	109	116	123	130	**140**	109
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	88	114	145	149	153	159	**156**	112
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	88	101	121	128	136	144	**155**	121
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	88	118	141	145	150	156	**154**	112
B5. Combination of B1–B4 using one–half standard deviation shocks	88	109	140	144	148	154	**151**	109
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	88	148	160	170	180	190	**205**	160

continued

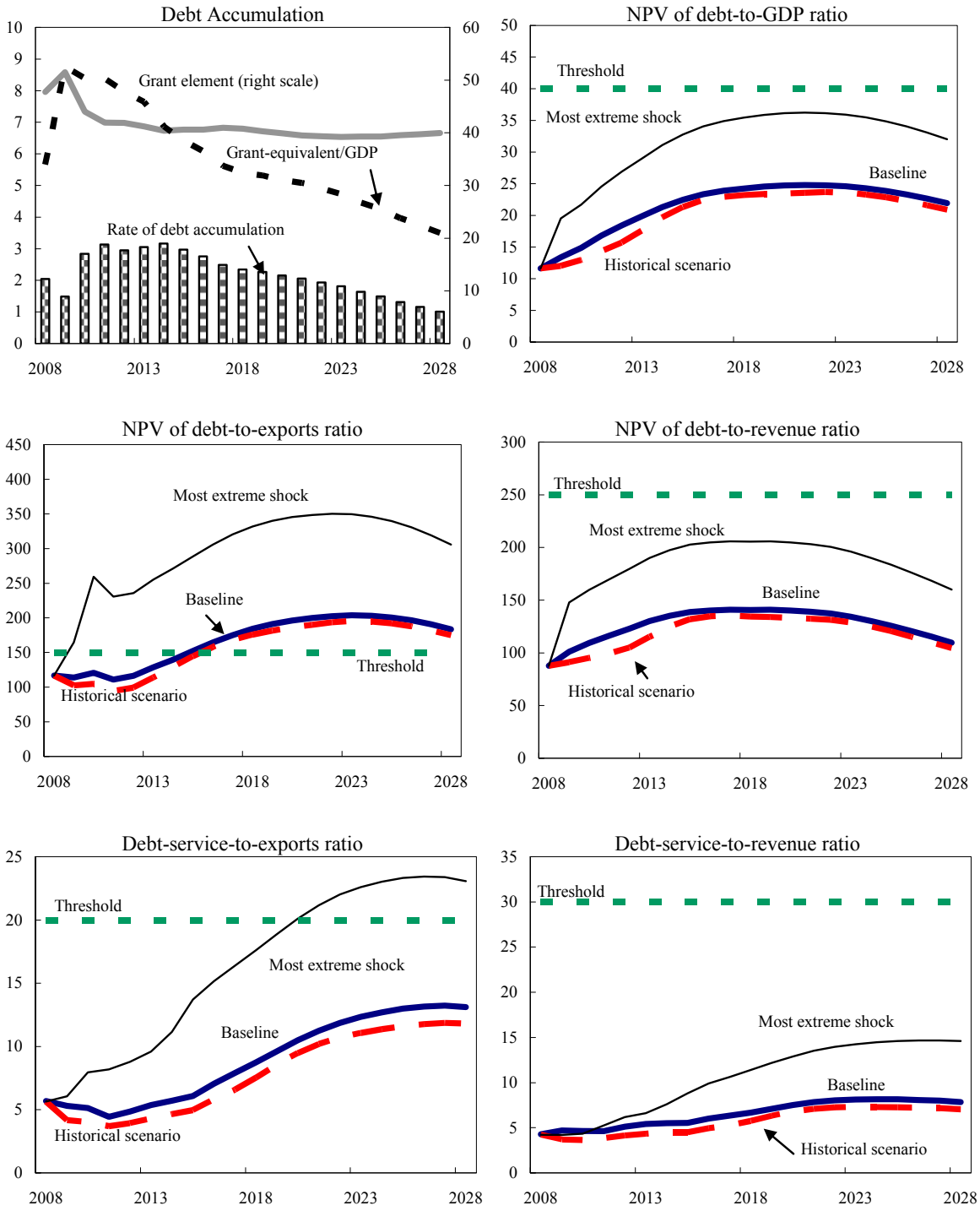
Appendix II—Table 2. Burkina Faso: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2008–28 (concluded)
(Percent)

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
Debt service-to-exports ratio								
Baseline	6	5	5	4	5	5	**9**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	6	4	4	4	4	4	**8**	12
A2. New public sector loans on less favorable terms in 2008–27 2/	6	5	5	5	6	7	**15**	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	6	5	5	4	5	5	**9**	13
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	6	6	8	8	9	10	**18**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	6	5	5	4	5	5	**9**	13
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	6	5	5	5	5	6	**11**	14
B5. Combination of B1–B4 using one–half standard deviation shocks	6	5	6	6	7	8	**14**	18
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	6	5	5	4	5	5	**9**	13
Debt service-to-revenue ratio								
Baseline	4	5	5	5	5	5	**7**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2008–27 1/	4	4	4	4	4	4	**6**	7
A2. New public sector loans on less favorable terms in 2008–27 2/	4	4	4	5	6	7	**11**	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2008–09	4	4	4	5	5	5	**7**	8
B2. Export value growth at historical average minus one standard deviation in 2008–09 3/	4	4	4	5	6	6	**8**	8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2008–09	4	4	5	5	6	6	**7**	9
B4. Net non–debt creating flows at historical average minus one standard deviation in 2008–09 4/	4	4	5	5	6	6	**8**	8
B5. Combination of B1–B4 using one–half standard deviation shocks	4	4	4	5	6	6	**8**	8
B6. One–time 30 percent nominal depreciation relative to the baseline in 2008 5/	4	6	6	7	7	8	**10**	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	34	34	34	34	34	34	**34**	34

Source: Staff projections and simulations.

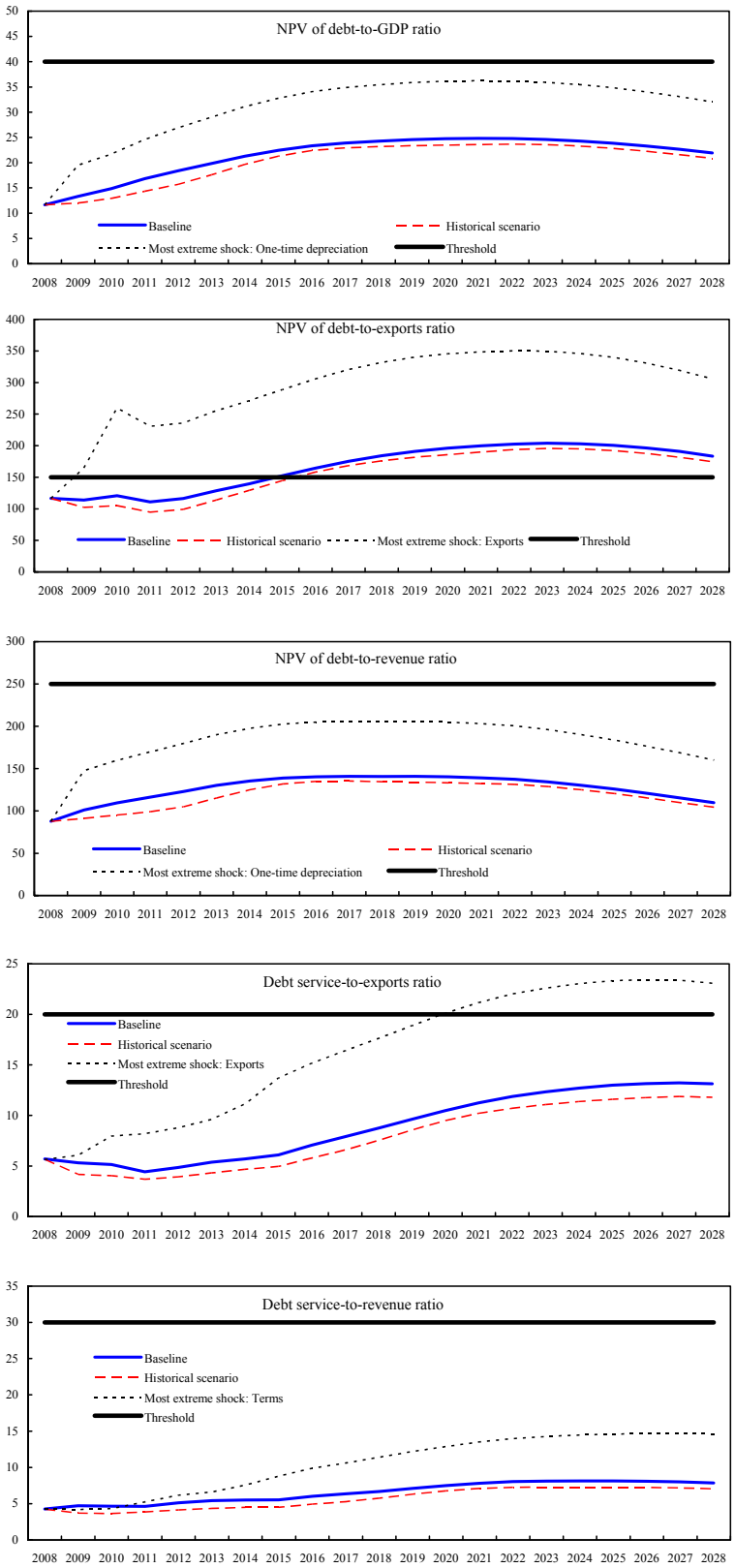
1/ Variables include real GDP growth, growth of GDP deflator (US$ terms), non–interest current account in percent of GDP, and non–debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assumingan offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Appendix II—Figure 3. Burkina Faso: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios (in percent), 2008-28 1/



Source: Staff projections and simulations.
1/ Thresholds for medium performer.

Appendix II—Figure 4. Burkina Faso: Indicators of Public External Debt in Alternative Scenarios, 2008–28 (Percent)



Source: IMF staff projections and simulations.

Appendix II—Table 3: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-28

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2005	2006	2007	Historical Average 5/	Standard Deviation 5/	2008	2009	2010	2011	2012	2013	2008-13 Average	2018	2028	2014-28 Average
Public sector debt 1/	44.4	22.8	22.0			23.7	25.1	28.0	30.3	31.6	32.8		36.6	32.1	
o/w foreign-currency denominated	40.9	20.0	18.6			21.4	20.1	22.2	24.4	26.3	28.0		33.2	25.4	
Change in public sector debt	0.3	-21.6	-0.8			1.7	1.4	2.9	2.3	1.4	1.2		0.4	-1.1	
Identified debt-creating flows	6.6	-23.0	2.4			3.1	4.8	4.0	2.1	1.6	1.3		0.4	-1.1	
Primary deficit	4.6	5.0	5.3	3.7	1.1	4.1	6.4	5.0	3.5	3.3	3.2	4.2	2.6	0.8	2.1
Revenue and grants	17.4	18.9	20.1			17.3	20.3	19.9	20.7	20.9	20.9		21.1	22.5	
of which: grants	4.6	5.9	6.5			4.0	7.1	6.4	6.3	5.9	5.6		3.8	2.5	
Primary (noninterest) expenditure	21.9	23.9	25.4			21.4	26.7	25.0	24.3	24.2	24.1		23.7	23.3	
Automatic debt dynamics	2.1	-5.9	-2.9			-1.0	-1.5	-1.0	-1.4	-1.6	-1.9		-2.2	-1.9	
Contribution from interest rate/growth differential	-3.6	-3.1	-1.0			-1.3	-0.7	-1.1	-1.4	-1.8	-1.9		-2.1	-1.9	
of which: contribution from average real interest rate	-0.7	-0.8	-0.2			-0.2	0.0	0.0	0.0	-0.1	-0.1		0.0	-0.1	
of which: contribution from real GDP growth	-2.9	-2.3	-0.8			-1.1	-0.7	-1.0	-1.4	-1.7	-1.8		-2.0	-1.8	
Contribution from real exchange rate depreciation	5.7	-2.7	-1.9			0.4	-0.8	0.0	0.0	0.1	0.0		...	...	
Other identified debt-creating flows	0.0	-22.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	-22.2	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-6.3	1.4	-3.1			-1.5	-3.4	-1.1	0.2	-0.2	-0.1		-0.1	0.0	
NPV of public sector debt	23.2	14.0	15.5			13.9	18.4	20.6	22.7	23.7	24.7		27.7	28.6	
o/w foreign-currency denominated	19.7	11.2	12.0			11.6	13.4	14.9	16.8	18.4	19.9		24.3	21.9	
o/w external	19.7	11.2	12.0			11.6	13.4	14.9	16.8	18.4	19.9		24.3	21.9	
NPV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Gross financing need 2/	6.0	6.2	6.6			4.8	7.2	5.8	4.4	4.2	4.1		3.9	2.4	
NPV of public sector debt-to-revenue and grants ratio (in percent)	133.6	73.9	77.0			80.6	90.7	103.3	109.3	113.5	118.1		131.3	127.1	
NPV of public sector debt-to-revenue ratio (in percent)	181.7	107.3	113.8			105.0	139.3	151.5	156.5	158.4	161.8		160.6	143.1	
o/w external 3/	154.2	86.1	88.5			87.7	101.2	109.4	116.3	123.1	130.2		140.7	109.6	
Debt service-to-revenue and grants ratio (in percent) 4/	8.2	6.2	6.7			4.0	3.9	4.1	4.1	4.4	4.7		5.9	7.2	
Debt service-to-revenue ratio (in percent) 4/	11.2	8.9	9.9			5.2	5.9	6.0	5.8	6.2	6.4		7.3	8.1	
Primary deficit that stabilizes the debt-to-GDP ratio	4.3	26.6	6.0			2.4	4.9	2.2	1.2	1.9	2.0		2.2	1.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.1	5.5	3.6	5.6	1.9	5.2	3.1	4.2	5.3	6.0	6.0	5.0	6.0	5.9	5.9
Average nominal interest rate on forex debt (in percent)	0.7	0.6	1.1	0.6	0.4	1.3	1.0	0.8	1.1	1.3	1.3	1.1	1.5	1.8	1.6
Average real interest rate on domestic currency debt (in percent)	3.9	5.1	3.9	-0.3	5.2	-2.7	5.3	1.9	1.1	0.9	1.0	1.2	1.2	-1.2	0.2
Real exchange rate depreciation (in percent, + indicates depreciation)	16.1	-7.3	-10.0	-2.6	12.5	2.2	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	0.6	-0.1	3.6	2.8	2.9	7.0	2.0	2.0	2.0	2.0	2.0	2.8	2.0	2.0	2.0
Growth of real primary spending (deflated by GDP deflator, in percent)	10.8	15.0	9.9	7.4	6.8	-11.3	28.6	-2.5	2.4	5.6	5.6	4.7	5.5	5.9	5.7
Grant element of new external borrowing (in percent)	0.0	0.0	0.0	0.0	0.0	47.7	51.5	44.0	41.9	41.9	41.2	44.7	40.8	40.0	...

Sources: Country authorities; and Fund staff estimates and projections.

1/ Central government.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

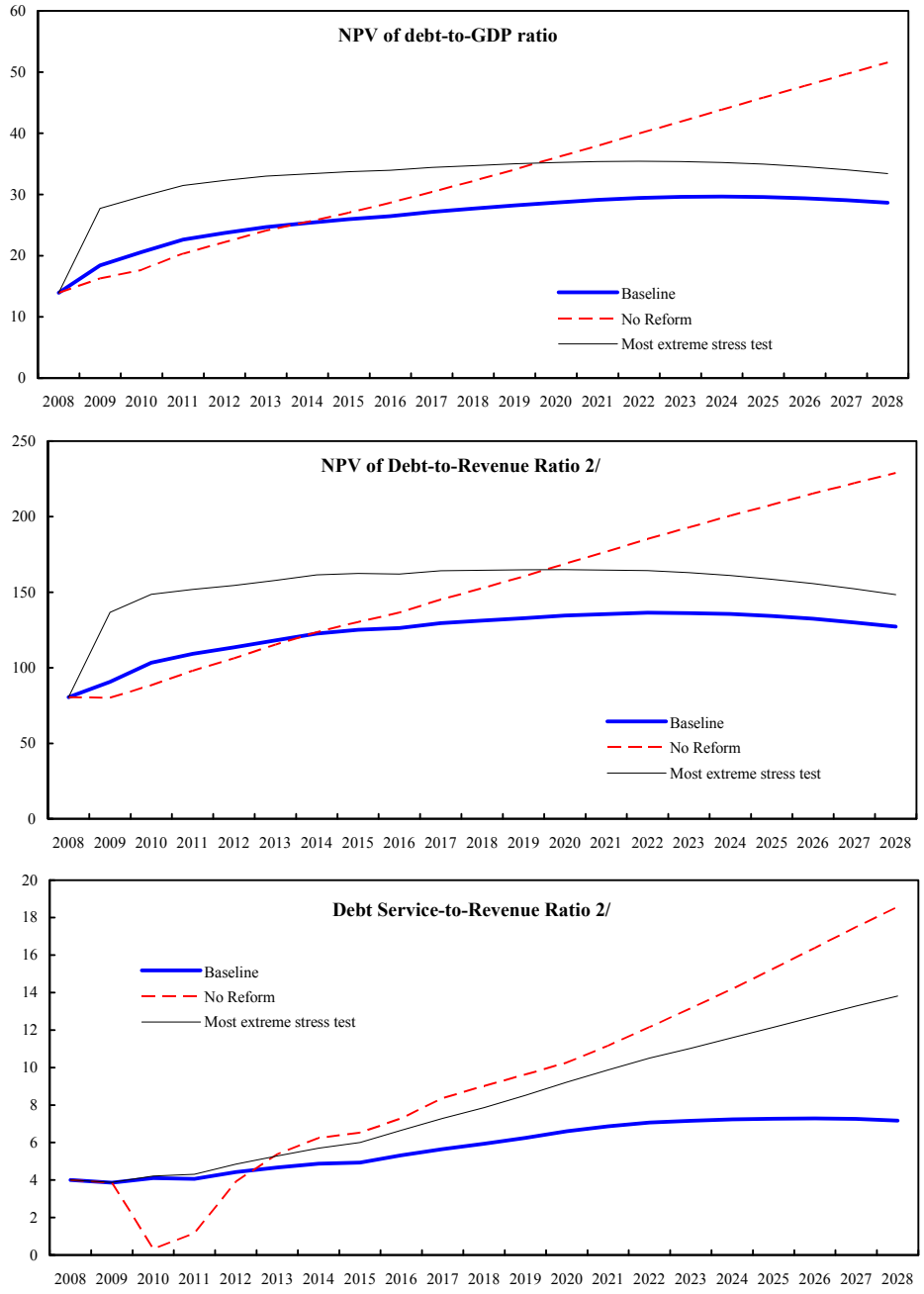
Appendix II—Table 4.Burkina Faso: Sensitivity Analysis for Key Indicators of Public Debt 2008-28

	Projections							
	2008	2009	2010	2011	2012	2013	2018	2028
NPV of Debt-to-GDP Ratio								
Baseline	14	18	21	23	24	25	28	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	14	16	16	19	20	22	29	47
A2. Primary balance is unchanged from 2008	14	16	18	20	22	24	32	52
A3. Permanently lower GDP growth 1/	14	19	21	23	25	26	32	43
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	14	18	20	23	24	25	27	28
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	14	17	19	21	22	23	26	28
B3. Combination of B1-B2 using one half standard deviation shocks	14	16	18	20	21	22	26	28
B4. One-time 30 percent real depreciation in 2009	14	24	25	27	27	28	29	31
B5. 10 percent of GDP increase in other debt-creating flows in 2009	14	28	30	31	32	33	35	33
NPV of Debt-to-Revenue Ratio 2/								
Baseline	81	91	103	109	113	118	131	127
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	81	78	84	92	99	107	139	209
A2. Primary balance is unchanged from 2008	81	80	89	98	106	115	153	229
A3. Permanently lower GDP growth 1/	81	91	105	112	117	124	150	190
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	81	90	103	109	113	117	130	125
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	81	83	95	101	106	111	125	123
B3. Combination of B1-B2 using one half standard deviation shocks	81	80	89	97	102	108	125	126
B4. One-time 30 percent real depreciation in 2009	81	117	127	130	132	134	139	140
B5. 10 percent of GDP increase in other debt-creating flows in 2009	81	137	149	152	154	158	165	148
Debt Service-to-Revenue Ratio 2/								
Baseline	4	4	4	4	4	5	6	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	4	4	0	1	3	4	8	16
A2. Primary balance is unchanged from 2008	4	4	0	1	4	5	9	19
A3. Permanently lower GDP growth 1/	4	4	4	4	5	5	8	14
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010	4	4	4	4	4	5	6	7
B2. Primary balance is at historical average minus one standard deviations in 2009-2010	4	4	1	3	4	4	6	7
B3. Combination of B1-B2 using one half standard deviation shocks	4	4	1	2	4	4	6	7
B4. One-time 30 percent real depreciation in 2009	4	4	4	5	5	6	7	9
B5. 10 percent of GDP increase in other debt-creating flows in 2009	4	4	21	10	8	7	8	11

Sources: Country authorities; and Fund staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Appendix II—Figure 5. Burkina Faso: Indicators of Public Debt Under Alternative Scenarios, 2008-28 1/



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2018.
2/ Revenue including grants.